Andrea Fossà

Master Student in artificial intelligence

Roncà, Italy   andrea.fossa1801@gmail.com   +39 3450778568   website   linkedin   github

Education

MS in Artificial Intelligence, University of Bologna (link) — *Sept 2023 – Mar 2026*

- Class: LM-18 Computer science, LM-32 Computer systems engineering
- **Key Coursework:** AI for Medicine (NER, Gene Ontology, Clinical Data), Natural Language Processing (LLMs, Transformers), Deep Learning, Computer Vision, Knowledge Representation & Reasoning (Ontologies, Semantic Web).
- **Thesis/Focus:** Application of Deep Learning models to biological data integration and analysis. GPA 106/110

BS in Bioinformatics, University of Verona (link) — *Sept 2020 – Jul 2023*

- Class: : L-31 Computer science
- GPA: 107/110

Experience

Consultant — *Padova, Italy*
Reply SPA (link) — *Apr 2026 – current*

AI Research Intern — *Trento, Italy*
Fondazione Bruno Kessler, FBK E3DA Lab (link) — *Mar 2025 – Sep 2025*

Internship focused on the processing of bio signals (EEG) using deep learning techniques on edge devices. Energy Efficient Embedded Digital Architectures unit (E3DA).

BCI Research Intern — *Verona, Italy*
BraiNavLab (link) — *Mar 2023 – Jul 2023*

Developed passive BCI for assessing Mental Workload

Stage — *Lonigo, Italy*
Fabbrica Italiana Sintetici (link) — *May 2018 – Jun 2018*

Technologies

Languages: Python, C, Java, Matlab, R.

Tools, Libraries & Skills: PyTorch, TensorFlow, Scikit-learn, OpenCV, Z3Py, Github, Docker, SQL, CUDA, AWS, Azure, Biomedical Ontologies, ChEMBL/PubChem (familiarity), HuggingFace, Spacy, Linux.

Bioinformatics: UniProt, PDB, BLAST, ClustalOmega, ITK-SNAP

Projects & publications

Explicit modelling of subject dependency in BCI decoding — *ArXiv link*

- Developed Deep Learning models for EEG signal processing using PyTorch.

Advanced NLP Pipeline for Unstructured Text Analysis — *github.repo*

- Developed an end-to-end NLP pipeline using Transformers (BERT/RoBERTa) and LLMs for semantic analysis and classification of noisy, unstructured text. Implemented rigorous data cleaning, tokenization, and text preprocessing workflows to improve model performance on raw data.Optimized model fine-tuning techniques, applicable to information extraction tasks from diverse sources (e.g., literature, reports).

AI for Medicine: Diagnostic & Genomics Analysis — *Academic Project*

- Implemented ML algorithms for medical imaging and genomic sequence analysis, integrating biological datasets.

Multi-Modal Walkable Population Prediction via Satellite Imagery & OSM Fusion *github.repo*

Semantic Segmentation of Out-of-Distribution Objects in Autonomous Vehicle Navigation *github.repo*

Satellite image semantic segmentation *github.repo*

From Scratch AI *github.repo*

Detecting Objects: Classic and NN

Combinatorial Decision Making and Optimization for Multiple Courier Planning problem *github.repo*

Languages & Soft Skills

Promote communication between people with different cultural backgrounds *link*

Objective: Sustainable Development *link*

Strategic problem solving patterns for better creativity *link*

Languages: Italian: native, English: fluent